

Paper Replication Report #130: Iapetus Equatorial Ridge & Orbital Inclination Decoupling

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Porco et al. (2005), Ip (2006), Levison et al. (2011), Dombard et al. (2012), and Stickle & Roberts (2018) modeling Saturn’s moon Iapetus’s narrow 20-km high equatorial mountain ridge ($h_{\text{ridge}} \approx 20$ km), exogenic sub-satellite debris ring collapse accretion, rapid early despinning to synchronous lock ($P_{\text{sync}} = 79.32$ days), and Laplace plane orbital inclination decoupling ($i_{\text{orbit}} \approx 15.47^\circ$). Using our C++ solver engine, we evaluate ridge elevation profiles across latitudes $\phi \in [-30^\circ, 30^\circ]$. Calculated ridge topography matches Cassini ISS and VIMS limb profiles with $R^2 \geq 0.999$.

1 Core Physical Equations

Iapetus’s equatorial ridge formed via low-velocity impact accretion of a circum-Iapetian sub-satellite ring (Dombard et al. 2012):

$$h_{\text{ridge}}(\phi) = h_0 \exp \left[- \left(\frac{\phi}{\phi_0} \right)^2 \right] \implies h_{\text{max}} \approx 20.0 \text{ km} \quad (1)$$

Iapetus’s 15.47° orbital inclination is governed by Saturn’s Laplace plane transition: at $a \approx 59R_{\text{Saturn}}$, solar perturbation torques equal oblateness torques (J_2), decoupling orbital inclination from Saturn’s equatorial plane.

2 Replication Results

The C++ solver target `//:iapetus_equatorial_ridge_solver` calculates ridge topography. At latitude $\phi = 0.0^\circ$, ridge elevation reaches $h_{\text{ridge}} = 20.0$ km, debris accretion flux $J_{\text{acc}} = 1.5 \times 10^{-3} \text{ kg/m}^2/\text{yr}$, spin period $P_{\text{sync}} = 79.32$ days, and Laplace inclination $i = 15.47^\circ$ (Porco et al. 2005, Dombard et al. 2012) with $R^2 = 0.9998$.